

Rising Waters

A Machine Learning Approach to Flood Prediction

Define Problem Statements

Phase 1 — Problem Definition

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1. Problem Statement

Floods are among the most devastating natural disasters, claiming thousands of lives and displacing millions of people every year. Despite their recurring nature, the lack of timely and accurate early-warning systems continues to amplify their destructive impact. Conventional forecasting methods often fall short in predicting floods at the right time, leaving authorities and communities with insufficient opportunity to respond.

2. Problem Decomposition

2.1 Root Causes

- Climate change and unpredictable rainfall patterns increasing flood frequency and severity
- Rapid urbanization and deforestation reducing natural water absorption
- Inadequate drainage and water management infrastructure
- Absence of data-driven, scalable early-warning mechanisms accessible to local authorities

2.2 Consequences

- Loss of human life and mass displacement of communities
- Destruction of homes, infrastructure, and agricultural land
- Significant economic losses and long-term environmental damage
- Overwhelmed disaster management systems with insufficient response time

3. Problem Statement (Formal)

How might we provide disaster management authorities and local communities in flood-prone regions with accurate, timely flood risk predictions using historical meteorological data, so that they can take proactive protective measures and minimize loss of life and property?

4. Scope Definition

In Scope	Out of Scope
Prediction using historical rainfall/weather data	Real-time IoT sensor integration
Binary classification: flood vs. no-flood	Severity level prediction (mild/severe/extreme)
Flask web application for data entry	Mobile application development
IBM Cloud / Render deployment	Integration with government alert systems
Model comparison and selection	Long-term time-series forecasting

5. Success Criteria

- Model achieves above 90% accuracy on test data
- Flask application correctly processes inputs and returns predictions

- System is accessible via a public URL (deployed to cloud)
- All five development Epics completed and reviewed in SkillWallet